

Does a WNBA Team Increase Girls' High School Sports Participation?

Evidence from a Staggered Difference-in-Differences Design

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Outline

- 1 Motivation & Research Question
- 2 Data
- 3 Empirical Strategy
- 4 Results
- 5 Conclusion
- 6 Appendix

- Title IX (1972) dramatically expanded female athletics, but **role-model effects** remain understudied.
- WNBA founded 1997 — first major professional women's basketball league in the US.
- Franchises entered states **staggered across 1997–2018**, creating quasi-experimental variation.
- **Question:** Does local exposure to a professional women's team *increase* girls' participation in high school sports?

Why it matters

- Physical/mental health benefits of youth sports
- Long-run labour market returns to athletic participation
- Informing policy on sports funding and media representation

Girls' Participation — NFHS

- National Federation of State High School Associations
- Annual PDF reports, scraped & digitised
- **Sports:** Basketball, Soccer, Cross Country, Track & Field
- **Coverage:** 51 states, 1993–94 through 2018–19 (26 years)

Treatment — WNBA Franchises

- Franchise history from official WNBA records
- Treatment: first year any franchise operates in a state
- **18 treated states**, 33 never-treated states
- 8 distinct treatment cohorts (1997–2010)

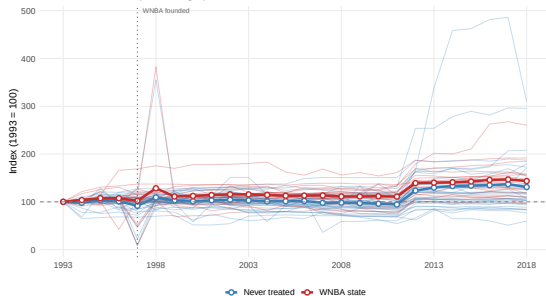
Table: Mean Girls' High School Participation by Sport (1993–2018)

Sport	Ever-Treated States	Never-Treated States
Basketball	15,725	5,741
Cross Country	7,022	2,162
Soccer	11,381	8,221
Track & Field	19,109	7,652
Number of States	18	33

Trends in Participation

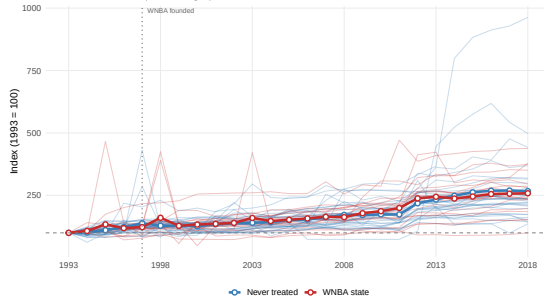
Girls' Basketball: Participation Index (1993 = 100)

Thin lines = individual states | Bold lines = group means



Girls' Cross Country: Participation Index (1993 = 100)

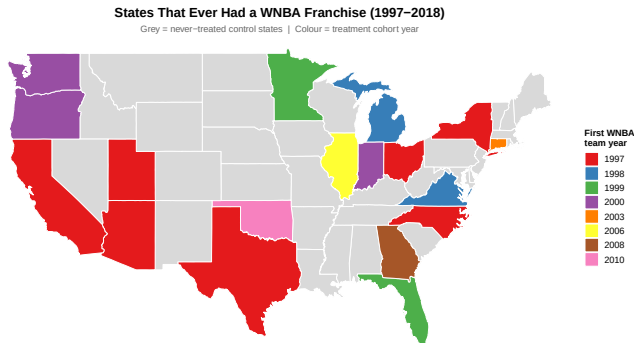
Thin lines = individual states | Bold lines = group means



Treatment Cohorts

Table: WNBA Franchise Entry by Cohort

Year	States	N
Never treated	—	33
1997	NY, TX, NC, AZ, OH, CA, UT	7
1998	VA, MI	2
1999	FL, MN	2
2000	OR, IN, WA	3
2003	CT	1
2006	IL	1
2008	GA	1
2010	OK	1
Total		18



Estimating equation

$$Y_{it} = \beta_0 + \beta_1 \mathbb{1}\{\text{WNBA}_{it}\} + \epsilon_{it}$$

- Y_{it} : number of girls participating in a given sport in state i and year t .
- $\mathbb{1}\{\text{WNBA}_{it}\}$: indicator for whether state i has a WNBA team in year t .
- ϵ_{it} : error term.
- Future versions will include state and year fixed effects, as well as covariates (e.g., population, median income).

This specification is biased with staggered treatment timing, so I correct with Callaway & Sant'Anna (2021) method.

Identification Strategy

Callaway & Sant'Anna (2021) staggered difference-in-differences:

- Define **group** g = first year state i gets a WNBA team (or $g = 0$ if never treated).
- Estimate **group-time average treatment effects**:

$$\widehat{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g]$$

- **Control group**: never-treated states only (33 states).
- Allows for staggered timing whereas baseline TWFE is biased
- Standard errors clustered by state; 95% uniform confidence bands.

Key assumption

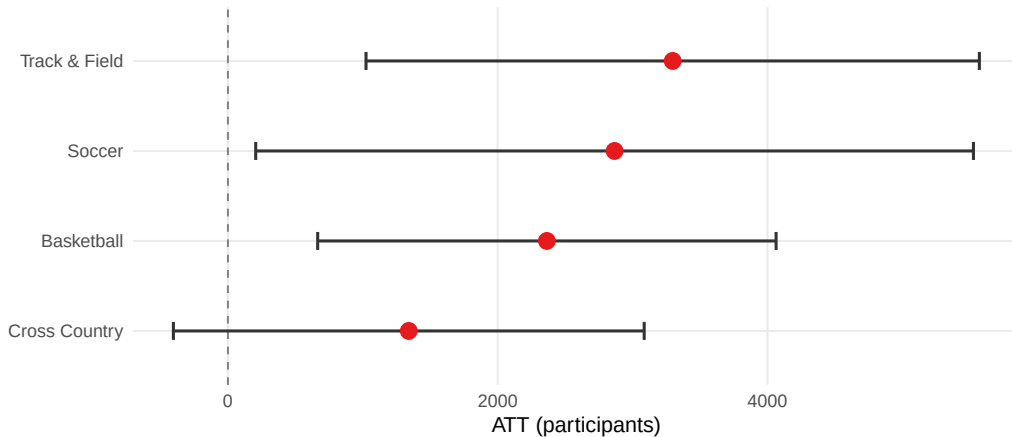
Parallel trends conditional on covariates: in the absence of a WNBA team, treated and never-treated states would have evolved similarly. We test this via pre-treatment event-study coefficients.

Table: C&S Simple-Aggregate ATT by Sport

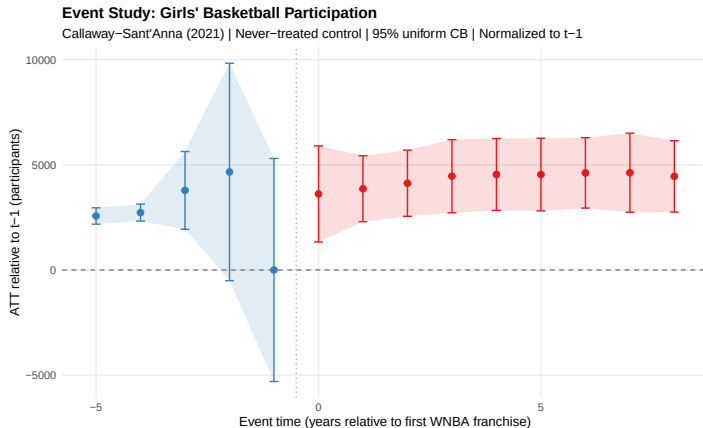
Sport	ATT	SE	<i>p</i>
Basketball	2365	867	0.006
Soccer	2867	1357	0.035
Cross Country	1341	890	0.132
Track & Field	3297	1160	0.005

Overall ATT: Effect of WNBA Franchise on Girls' Participation

Callaway–Sant'Anna (2021) | Never–treated control | 95% CI



Event Study: Basketball



- Pre-period coefficients **centred near zero** — supports parallel trends.
- Post-entry: No discernable effect until year 4, though trending positively.
- Effect slowly grows after initial entry before seeming to stabilize at an increase of about 2000 additional girls.

Event Studies: All Sports

Event Study: Girls' Soccer Participation

Callaway-Sant'Anna (2021) | Never-treated control | 95% uniform CB | Normalized to t-1

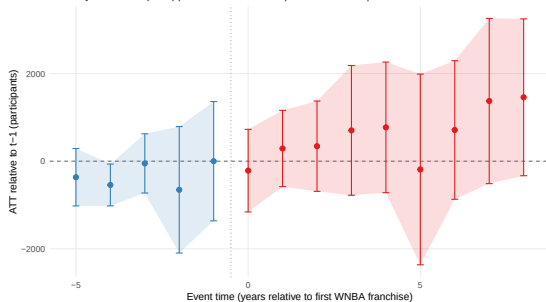


Figure: Soccer

Event Study: Girls' Cross Country Participation

Callaway-Sant'Anna (2021) | Never-treated control | 95% uniform CB | Normalized to t-1

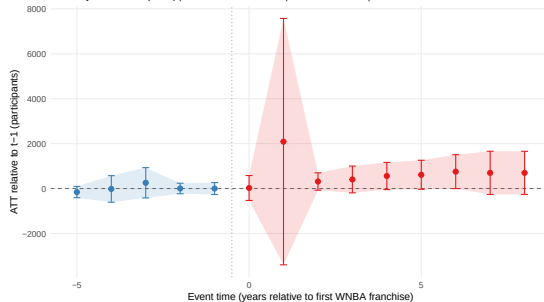


Figure: Cross Country

Event Study: Girls' Track & Field Participation

Callaway-Sant'Anna (2021) | Never-treated control | 95% uniform CB | Normalized to t-1

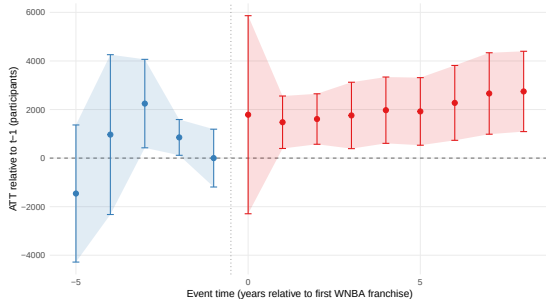
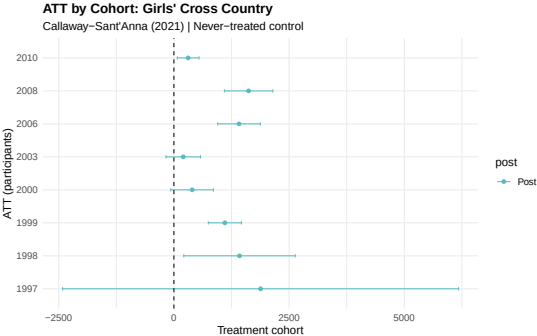
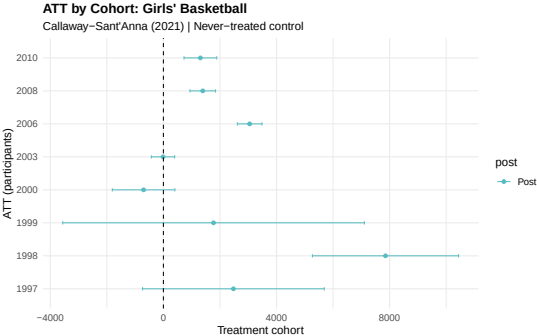


Figure: Track & Field

- Soccer: No evidence of any effect; coefficients are small with massive confidence intervals.
- Cross Country: Positive, but not statistically significant; wide confidence bands.
- Track & Field: Some evidence of a positive effect in early post-treatment years, but statistical significance is borderline.

ATT by Treatment Cohort: Basketball



Priority Items & Limitations

- NFHS data digitised from annual PDF reports — potential for transcription errors.
- WNBA treatment defined at state level; may not capture local variation in media exposure or fan engagement.
- Baseline TWFE estimates are biased with staggered treatment timing; C&S method addresses this but relies on parallel trends assumption.
- Event-study coefficients have wide confidence intervals, especially for non-basketball sports, limiting precision of inference.
- Add state and year covariates (e.g., population, median income) to improve precision and test robustness.

Main findings

- Pre-trends pass visual inspection across all sports.
- WNBA entry associated with a modest, but significant increase in basketball participation (approx. 2000 additional girls)
- No significant effects on soccer or cross country; some suggestive evidence for track & field seeing a boost, but statistical significance is borderline.

Economic implications

Role-model effects can boost youth sports participation, with potential long-term health and economic benefits.

Next steps

- Synthetic Difference-in-Differences
- Explore substitution vs. complementarity across sports (e.g., does WNBA entry boost basketball but reduce soccer?)

Thank you

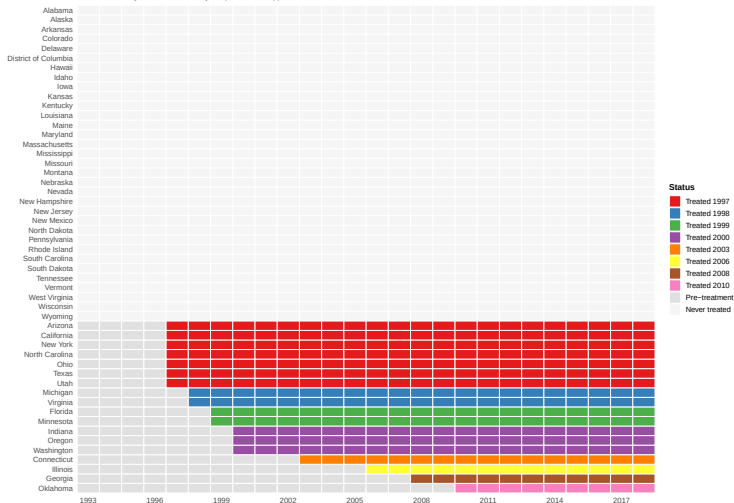
Questions welcome

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Appendix: Additional Figures

WNBA Franchise Entry: Staggered Treatment Panel

States sorted by treatment cohort year (earliest at top)



Appendix: Additional Figures, cont.

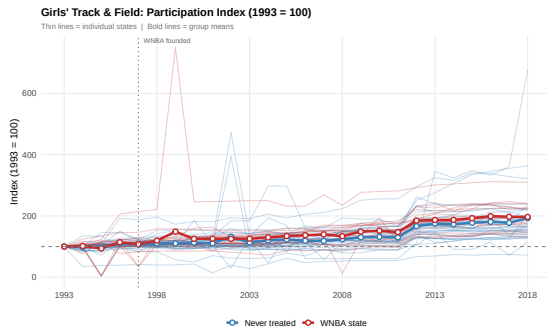


Figure: Spaghetti Plot of Track & Field Participation (Relative to 1993 Baseline)

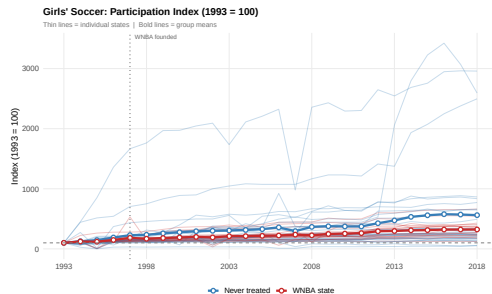


Figure: Spaghetti Plot of Soccer Participation (Relative to 1993 Baseline)